

KerNet: A Novel Deep Learning Approach for Keratoconus and Sub-Clinical Keratoconus Detection Based on Raw Data of the Pentacam HR System

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Abstract—Keratoconus is one of the most severe corneal diseases, which is difficult to detect at the early stage (i.e., sub-clinical keratoconus) and possibly results in vision loss. In this paper, we propose a novel end-to-end deep learning approach, called KerNet, which processes the raw data of the Pentacam HR system (consisting of five numerical matrices) to detect keratoconus and sub-clinical keratoconus. Specifically, we propose a novel convolutional neural network, called KerNet, containing five branches as the backbone with a multi-level fusion architecture. The five branches receive five matrices separately and capture effectively the features of different matrices by several cascaded residual blocks. The multi-level fusion architecture (i.e., low-level fusion and high-level fusion) moderately takes into account the correlation among five slices and fuses the extracted features for better prediction. Experimental results show that: (1) our novel approach outperforms state-of-the-art methods on an in-house dataset,

by $\sim 1\%$ for keratoconus detection accuracy and ~ 4 for sub-clinical keratoconus detection accuracy; (2) the attention maps visualized by Grad-CAM show that our KerNet places more attention on the inferior temporal part for sub-clinical keratoconus, which has been proved as the identifying regions for ophthalmologists to detect sub-clinical keratoconus in previous clinical studies. To our best knowledge, we are the first to propose an end-to-end deep learning approach utilizing raw data obtained by the Pentacam HR system for keratoconus and subclinical keratoconus detection. Further, the prediction performance and the clinical significance of our KerNet are well evaluated and proved by two clinical experts. Our code is available at <https://github.com/upzheng/Keratoconus>.

Index Terms—Keratoconus, sub-clinical keratoconus, Pentacam HR system, raw data, end-to-end deep learning.

I. INTRODUCTION

KERATOCONUS is one of the most severe corneal diseases in the world, which may result in irregular astigmatism, corneal edema, and eventually severe vision loss [1]. Early detection and intervention can alleviate the burden of treatment costs and even prevent keratoconus patients from vision loss. Clinically, keratoconus can be characterized by corneal dilatation, central thinning, and conical shape. Ophthalmologists generally diagnose such diseases based on corneal topography and tomography, which describe the surface curvature of the cornea [2].

The Pentacam HR system is one of the reliable commonly-used tools to assist ophthalmologists detecting keratoconus. It consists of a rotating Scheimpflug camera and a Pentacam software. The camera is used to collect corneal 3D geometric data of the eyes, and the software analyzes and visualizes the data for ophthalmologists. In this manner, the Pentacam HR system efficiently and visually offers ophthalmologists a series of calculated parameters (thousands of dimensions of normalized indices), six corneal topographic maps (for curvature, elevation, and pachymetry), and a corneal 3D image. Hence, an experienced ophthalmologist may detect keratoconus based on the visualized information provided by the Pentacam HR system. However, such a procedure for Pentacam-aided keratoconus

Manuscript received October 14, 2020; revised February 24, 2021 and April 16, 2021; accepted May 8, 2021. Date of publication May 12, 2021; date of current version October 5, 2021. This research was partially supported in part by the National Research and Development Program of China under Grant 2019YFB1404802, in part by the National Natural Science Foundation of China under Grant 61672453, in part by the Zhejiang University Education Foundation under Grants K18-511120-004, and K17-511120-017, and K17-518051-02, in part by the Zhejiang public welfare technology research project under Grant LGF20F020013, in part by the Medical and Health Research Project of Zhejiang Province of China under Grant 2019KY667, in part by the Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang under Grant 019R01007, in part by the Wenzhou Bureau of Science and Technology of China under Grant Y2020082, and in part the Key Laboratory of Medical Neurobiology of Zhejiang Province. The work of Danny Z. Chen was supported in part by NSF Grant CCF-1617735. (Ruiwei Feng, Zhe Xu, and Xiangshang Zheng contributed equally to this work.) (Corresponding authors: Jian Wu; Ke Yao.)

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Digital Object Identifier 10.1109/JBHI.2021.3079430

detection is generally time-consuming and the detection accuracy is experience-dependent. Furthermore, it is quite difficult for ophthalmologists to discriminate a patient with sub-clinical keratoconus from a healthy person [3], since the normalized indices and visualized corneal topographic maps might incur some data missing and thus indicate little obvious characteristics of the sub-clinical keratoconus eyes.

To address these issues, many machine learning methods have been proposed to help ophthalmologists detect keratoconus based on the Pentacam HR system in a more efficient way. However, most of these methods were used as a substitute for an ophthalmologist in the aforementioned detection procedure. That is, these methods generally received either the corneal topographic maps [4] or the normalized indices [2] generated by the Pentacam software for processing and modeling. In this manner, the Pentacam software played an important role of strong pre-processing. Although this software enabled both ophthalmologists and machine learning models to obtain lower dimensional data in an intuitive fashion, there are certain key drawbacks. For instance, it was found [3] that neither ophthalmologists nor the known machine learning methods could detect sub-clinical keratoconus effectively based on the above-mentioned processed data, possibly due to neglecting some information during the procedure of data processing by the Pentacam software. Yet, the neglected information might be recognizable for the detection of sub-clinical keratoconus. Hence, we think that more complete and primitive data without being processed by the Pentacam software should be utilized to detect sub-clinical keratoconus, as well as improving the accuracy of keratoconus detection. To our best knowledge, few studies were conducted to detect sub-clinical keratoconus using a deep learning approach and analyzing the raw data of the Pentacam HR system directly.

In this paper, we aim to fill the gap of utilizing the raw data collected by the Pentacam HR system for keratoconus and sub-clinical keratoconus detection. Specifically, the raw data derived by the Pentacam HR system is three CSV files containing five matrices, corresponding to the corneal 3D geometric data obtained by the Pentacam camera for the front and back surface curvature, the front and back surface elevation, and the pachymetry of the eyes. Each matrix has a size 141×141 and the nonzero elements in each matrix appear as the shape of the cornea. Empirically, there should be some implicit association among these five matrices, yet the association is probably not as strong as that among the channels in an RGB image. Thus, instead of imprudently regarding the five matrices as a whole, we originatively treat the five matrices as five image slices. In this view, a specific model is expected to take the five matrices respectively as input to extract features and then fuse the captured features in some way for better prediction.

We propose a novel deep learning approach, called KerNet, which is carefully designed to accommodate the specific format of the raw data by introducing the idea of multi-level fusion. Our KerNet consists of five branches, each containing four convolutional blocks with the residual structure in ResNet [5] and receiving an image slice as input. We introduce the idea of multi-level fusion to fuse the captured features of each slice. The multi-level fusion architecture is implemented by both

low-level fusion and high-level fusion, which are two main ways of multi-modal fusion [6]. For the low-level fusion, we build crossed connection between neighboring branches for feature fusion, followed by a specific spatial attention module. For the high-level fusion, the generated feature maps of the five branches are concatenated together channel-wise for final prediction, as illustrated in Fig. 1. Our main contributions are as follows.

- First, to our best knowledge, we propose the first end-to-end deep learning network utilizing the raw data obtained by the Pentacam HR system for keratoconus and sub-clinical keratoconus detection, instead of utilizing the data processed by the Pentacam software as in the previous methods [2], [4].
- Second, we propose a novel structure, KerNet, which receives raw data derived by the Pentacam HR system for feature extraction and final prediction, and exquisitely treats the data as five image slices. In this manner, our KerNet offers a new way to apply deep learning models for ophthalmic disease detection using data of a similar format as the raw data used in our study.
- Finally, the experimental evaluations demonstrate: (1) our KerNet can effectively detect sub-clinical keratoconus, which is a difficult task for both ophthalmologists and the previous machine learning methods [3] based on processed data; (2) our KerNet outperforms several well-known deep learning networks and achieves state-of-the-art performance for both keratoconus and sub-clinical keratoconus detection; (3) the performance of our KerNet has been verified and its clinical significance has been confirmed by two independent experts.

The rest of this paper is organized as follows. In Section II, we review the related work on keratoconus and sub-clinical keratoconus detection based on the aforementioned processed data, as well as the development of deep learning methods and their applications in ophthalmology. In Section III, we present the data utilized in this study (including the collection procedure and the distribution of the dataset) and the structure of our KerNet. Next, we conduct abundant experiments to verify the effectiveness of our KerNet in Section IV. Finally, we conclude this paper in Section V.

II. RELATED WORK

A. Pentacam-Aided Keratoconus Detection

Recently, many studies were conducted for keratoconus detection using the Pentacam HR system. Most of these researches were based on the processed data by the Pentacam software (e.g., calculated indices or visualized images), and few were based on raw data, as mentioned in Section I. These methods can be roughly divided into three groups according to the different formats of the data they used.

- 1) A lot of work has been done based on the calculated indices or even part of such indices [7]–[12]. These researches usually used basic statistical analysis methods [13]–[17] or traditional machine learning models [3], [18]–[22] to distinguish keratoconus (including sub-clinical keratoconus) from the normal eyes, with a limited

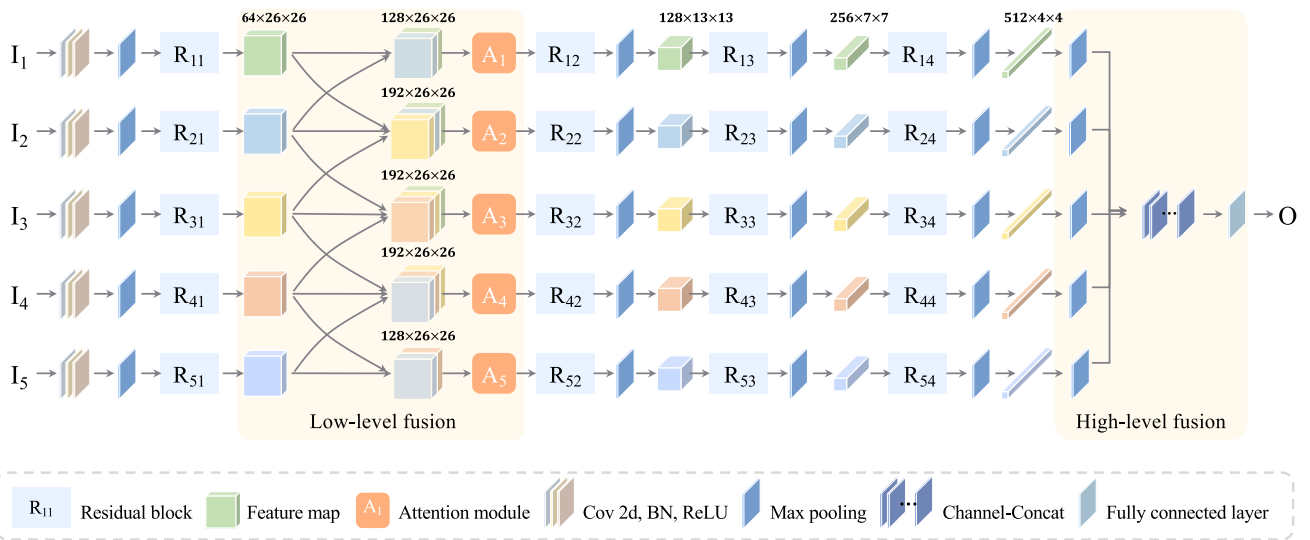


Fig. 1. Illustrating our KerNet model. I_i ($i \in \{1, 2, 3, 4, 5\}$) denotes different input slices and O denotes the output prediction. We highlight the operations of the low-level fusion and high-level fusion in our KerNet with a yellow background. R_{ij} ($i \in \{1, 2, 3, 4, 5\}, j \in \{1, 2, 3, 4\}$) represents the residual blocks and A_i ($i \in \{1, 2, 3, 4, 5\}$) represents the attention blocks. Different subscripts are utilized to show the diversity of the parameters in different blocks or modules. Besides, the output feature maps of each block or module are shown using different color cubes, with their sizes illustrated ($C \times W \times H$, where C is the channel size of a feature map).

number of calculated parameters obtained by the Pentacam HR system. For instance, the performance of a support vector machine algorithm was evaluated using 22 selected parameters [21]. Logistic regression algorithms were applied to detect the cases of sub-clinical and definite keratoconus [18], [22]. Studies were conducted based on 55 parameters [3] and 28 parameters [19]. Recently, Hallett *et al.* [2] proposed a deep learning method based on the whole calculated parameters (the Pentacam progress index (PPI)) for keratoconus classification.

- 2) A small number of studies were based on topography images visualized by the Pentacam HR system [23]. Toutounchian *et al.* [4] proposed an artificial intelligence algorithm to identify keratoconus by employing features of topographical maps obtained by the Pentacam HR system. Lavric *et al.* [14] proposed a convolutional neural network (CNN) to distinguish keratoconus from the normal eyes using a dataset of 3000 corneal topography images. This work was considered as the first to integrate a convolutional neural network into the diagnostic process of keratoconus. Besides, Kovacs *et al.* [13] measured tomographic data, topographic data, and keratoconus indices based on the Pentacam HR system. The authors in [24] proposed the PIRSS model based on corneal tomographic images to detect potential candidates for refractive surgery.
- 3) Very few studies were conducted based on the raw data obtained by the Pentacam HR system. In [25], a custom-made mathematical model was proposed for parameter selection from the raw data, and the selected parameters were utilized to detect clinical and suspected keratoconus by a feed-forward neural network (FFN). Although this work used raw data produced by the Pentacam HR system, it was not an end-to-end method for keratoconus detection.

B. Deep Learning Methods and Their Applications in Ophthalmology

Many deep learning methods have been developed for image recognition, since Krizhevsky *et al.* [26] proposed AlexNet in the 2012 ImageNet competition. Subsequently, VGGNet [27], GoogLeNet [28], ResNet [5], DenseNet [29], and other convolutional neural networks (CNNs) were proposed for image recognition in many different scenarios. Recently, deep learning has become one of the powerful tools for disease detection and medical information classification [30]. In ophthalmology, deep learning methods have been applied widely to diagnose and screen for diseases, such as diabetic retinopathy (DR) [31], age-related macular degeneration (AMD) [32], keratoconus [14], cataracts [33], *et al.* [34], as reviewed in [35]. As for keratoconus, many studies used deep learning methods for keratoconus detection and severity grading based on various datasets [36] (not only data obtained by the Pentacam HR system). Here we review the work using deep learning methods based on other assistant systems. Kamiya *et al.* [37] applied deep learning methods to discriminate different grades of keratoconus from the normal eyes based on six color-coded maps measured by an anterior segment optical coherence tomography. Dos Santos *et al.* [38] proposed a deep learning method, CorneaNet, to segment cornea OCT scans and detect keratoconic eyes, based on a custom-built ultrahigh-resolution OCT system. Several deep learning methods for keratoconus detection were based on the Artemis-1n system [39], EyeSys [40], and CASIA ASOCT [37]. Besides, in [41], ResNet was utilized for keratoconus detection based on the raw numerical data of the Orbscan system, which is quite similar to the data used in this paper. However, for detecting keratoconus and sub-clinical keratoconus, end-to-end deep learning methods based on the raw data derived by the Pentacam HR system are still lacking.

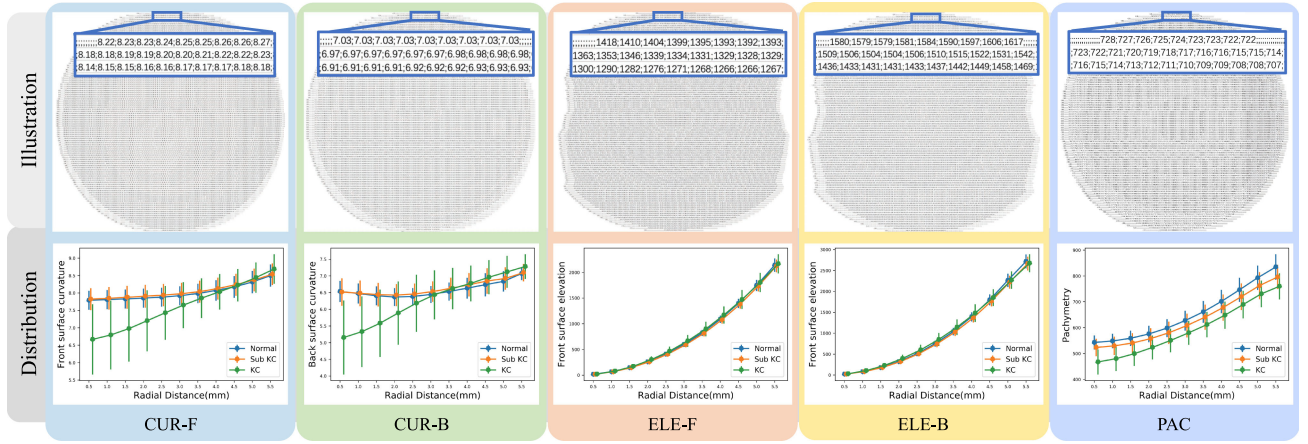


Fig. 2. Illustrating the raw data used in this study: An example of an eye with keratoconus. The top portion: five numerical matrices of size 141×141 each (from left to right: front surface curvature, back surface curvature, front surface elevation, back surface elevation, and pachymetry). The bottom portion: five figures of data distributions corresponding to the five slices, plotted with the radius of r (in $[0.5\text{mm}, 5.5\text{mm}]$) on the x -axis and the mean and standard deviation values on the y -axis. In each such figure, the curves of different colors represent data in different groups (keratoconus: green; sub-clinical keratoconus: orange; normal: blue).

III. MATERIAL AND METHODS

Fig. 1 gives an overview of our KerNet, a novel end-to-end deep learning network for keratoconus and sub-clinical keratoconus detection. Using the raw data (five matrices for each sample) exported from the Pentacam HR system, KerNet classifies the samples into three categories: normal, sub-clinical keratoconus, and keratoconus. In this section, the collection details and distribution of our dataset are first described. Then, the detailed structure of our KerNet is presented.

A. Dataset

1) Data Collection: In this study, we acquire a corneal dataset using the Pentacam HR system (Oculus, GmbH, Wetzlar, Germany). Our dataset contains 854 samples, including 335 normal samples (85 women and 250 men; 24.3 ± 2.4 years), 172 sub-clinical keratoconus samples (45 women and 124 men; 22.3 ± 5.6 years), and 347 keratoconus samples (90 women and 237 men; 24.2 ± 5.7 years). Only one eye of each patient and subject was included. The spherical equivalent (SE, Diopter) results of normal samples (-4.96 ± 2.23 D) and sub-clinical keratoconus samples (-4.80 ± 2.70 D) were different from keratoconus samples (-7.37 ± 4.54 D, analysis of variance ANOVA, $P < 0.05$). The best corrected visual acuity (BCVA, decimal VA) results of normal samples (1.0 ± 0.1 D) and sub-clinical keratoconus samples (1.0 ± 0.1 D) were different from keratoconus samples (0.5 ± 0.2 D, analysis of variance ANOVA, $P < 0.05$). The aforementioned statistical analyses were performed by the Statistical Package for the Social Sciences software (ver. 17, SPSS, Inc., Chicago, IL, USA). Comparisons among the normal, sub-clinical keratoconus, and keratoconus groups were made using ANOVA. $P < 0.05$ was defined as the level of statistical significance.

The corneal tomographic examination and data collection procedure is as follows. First, a subject is required to rest his/her chin on the chin rest of the examination instrument with the

forehead against the forehead band. An operator holds the lever and adjusts the lever according to the direction of the screen. Before the examining operation, the subject is required to fully blink his/her eyes so that the cornea is covered with a smooth film of tears. Then, in order to achieve acceptable examination results, the subject needs to keep the eyes open after the blue light scan for 2 seconds until the “Examination Quality Specification” shows “OK”. After examination, the corneal information (curvature, elevation, and pachymetry) of a subject is collected. Instead of using the Pentacam software for feature analysis or selection for final decision, we derive the raw data (three CSV files including five matrices for each sample) by a batch export functionality. This study was approved by the IRB of the 2nd Affiliated Hospital, School of Medicine, Zhejiang University.

2) Data Description: For each sample, the exported corneal data consists of five numerical matrices, each of a size 141×141 (corresponding to a radius of 7 mm). In this work, we treat the five numerical matrices as five 2D image slices, denoted as CUR-F, CUR-B, ELE-F, ELE-B, and PAC, corresponding to the front and back surface curvatures, the front and back surface elevations, and the pachymetry of an eye. **Fig. 2** gives an example of keratoconus with five slices. Then, we conduct statistical analysis for the five slices in three groups (normal, sub-clinical keratoconus, and keratoconus), respectively. It is worth noting that, due to the cover of eyelashes, eyelids, and so on, only the central part of each matrix thus obtained is authentic (around 5 mm) [42], while the data beyond 5 mm may be somewhat missing. Hence, only the data within 5.5 mm is analyzed, following the operations in [43]. Specifically, taking the CUR-F matrices of the normal group as an example, we compute the mean and standard deviation values of the elements on a circle with a radius of r . The values of r range from 0.5 to 5.5 mm, at intervals of 0.5 mm. That is, there are 11 values of r , yielding 11 mean and standard deviation values. The same operations are conducted for the matrices of CUR-B, ELE-F, ELE-B, and PAC in all the three groups, to form five sub-figures in **Fig. 2**.

By examining Fig. 2, one may first observe that the samples of keratoconus appear to contain some obvious characteristics in the CUR-F and CUR-B type samples. Especially, in the central position (within 3 mm), the mean and standard deviation values of the keratoconus samples are clearly lower than those of the sub-clinical keratoconus and normal samples. The second observation from Fig. 2 is that, all the three curves of ELE-F are quite close to one another, and the same is for ELE-B. This indicates that, statistically, the samples in the three different groups are hard to distinguish based on the matrices of ELE-F and ELE-B. Besides, it is noticeable that, in the five sub-figures, in all the sampled positions (sampled circles with different r values), the mean and standard deviation values of the sub-clinical keratoconus samples are quite similar to those of the normal samples. That is, the distribution of the sub-clinical keratoconus samples shows no significantly different characteristics. This seems to indicate that detecting sub-clinical keratoconus samples among other samples is challenging, especially among the normal samples.

B. Kernet

One can imagine that, for each sample, the five slices (CUR-F, CUR-B, ELE-F, ELE-B, and PAC) are somehow related to one another, since they are obtained by the same examination operations on the same eye. Nevertheless, considering the fact that each slice contains information of the same eye from diverse perspectives, we argue that the correlation among the five slices is likely not as strong as that among different channels of an RGB image. Accordingly, our KerNet is designed to moderately take into account such correlation among the five slices, while extracting information from each slice. This novel structure consists of five branches, each of which takes one slice as input. Furthermore, we introduce the idea of multi-level fusion based on a multi-modal fusion mechanism (i.e., both low-level fusion and high-level fusion), in order to fuse the extracted features of different slices. Fig. 1 shows an overview of our KerNet.

1) Backbone Structure: Our KerNet consists of five branches to receive five image slices respectively as input, instead of simply taking them as a whole five-channel input. The task of each branch is to process an input slice and extract information from it. The branch architecture consists of a general convolutional layer (followed by a batch normalization layer and a ReLU activation function), and four basic residual blocks in sequence, as illustrated in Fig. 1. Each residual block has two basic building blocks of residual connection, as in ResNet [5]. Given an input x , the output features of a building block are denoted as $ReLU(F(x) + x)$ [5]. The function $F(x)$ is implemented by two convolutional layers, each followed by a batch normalization layer (see Fig. 3). After each residual block, we conduct the down-sampling operation by max pooling. We opt for the residual block structure in KerNet due to (1) its well-designed structure for feature extraction [5] and (2) its broad applications and excellent performance in other image recognition tasks [44].

Specifically, we adapt each residual block based on the size of the input features. In this work, all the matrices (the input

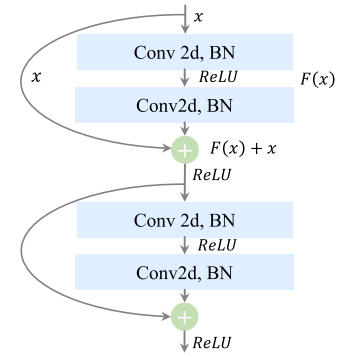


Fig. 3. The detailed structure of a residual block.

slices) are cropped at their central region of size 101×101 , corresponding to about $5 \text{ mm} \times 5 \text{ mm}$, for two considerations: 1) prior medical knowledge of clinical disease (keratoconus and sub-clinical keratoconus) detection, and 2) the data distribution of the dataset used in this research. Note that many previous studies (e.g., [42], [45]) have proved that corneal data in the central region of size $5 \text{ mm} \times 5 \text{ mm}$ is clinically significant for detecting keratoconus and sub-clinical keratoconus. Also, the statistical analysis in Section III-A2 indicates that the data beyond $5 \text{ mm} \times 5 \text{ mm}$ is somewhat missing. For our model to be more reliable, only the most truthful data is utilized for model training and evaluation. In this manner, given an input slice I_i , the output feature maps of the first residual block in each branch (denoted as R_{i1} for branch i) are of size $64 \times 26 \times 26$. More details are illustrated in Fig. 1.

2) Low-Level Fusion and High-Level Fusion: The architecture of our multi-level fusion (low-level fusion and high-level fusion) plays an important role in our KerNet. Specifically, we conduct low-level fusion after the first residual blocks and high-level fusion before the output layer, as highlighted in Fig. 1. We conduct these fusion operations for two considerations. On one hand, commonly, the low-level features captured by a CNN provide more spatial information and the high-level features provide more semantic information [46]. We argue that both the spatial features and the semantic features extracted in different branches (from different input slices) should be fused for better prediction. On the other hand, we suspect that too many fusing operations (e.g., fusing after each residual block) could be redundant. The effectiveness of these two considerations and such a structure is verified in Section IV.

For the low-level fusion, we build crossed connections between the neighboring branches for feature fusion, and introduce an attention mechanism for better fusion using a specific spatial attention module. Specifically, we take the feature map of the 2nd branch as an example for explanation. Given the input slice of this branch, I_2 , of size $1 \times 101 \times 101$, after the general convolutional layer and the first residual block, the feature map is of size $64 \times 26 \times 26$. Then, this feature map is concatenated with those of the two neighboring branches (the 1st and 3rd branches) in channel dimension, generating a new feature map of size $192 \times 26 \times 26$. The same concatenation operations are performed on the other four branches. Note that it is different

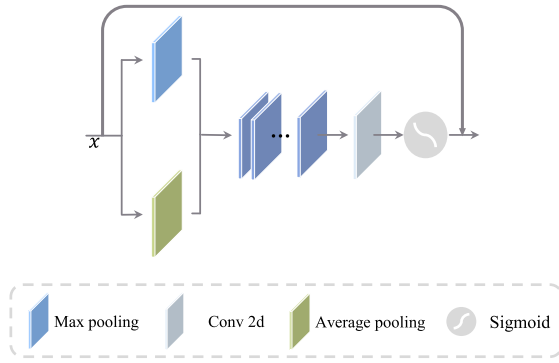


Fig. 4. The detailed structure of a spatial attention module.

that both the feature maps of the 1st and 5th branches after concatenation are of size $128 \times 26 \times 26$, while those of the other three branches are of size $192 \times 26 \times 26$ (as shown in Fig. 1). With the concatenated feature maps, further feature selection and better fusion are conducted by introducing spatial attention modules [47]. Such spatial attention modules have been well studied [47], and we use them to guide KerNet on where to pay more attention. We introduce such attention modules due to the clinical fact that ophthalmologists generally pay more attention to certain regions, and not the whole area of an image slice is of the same importance for keratoconus recognition.

As shown in Fig. 4, for an input x , the spatial attention module first conducts both global max pooling and global average pooling to aggregate the channel features of x . Then, the generated feature maps are concatenated to form the spatial-attention map M_x after a convolution layer and a Sigmoid function, which can be computed as:

$$M_x = \frac{1}{1 + \exp(-W \cdot C(\{G_a(x), G_m(x)\}))} \odot x \quad (1)$$

where G_a is global average pooling, G_m is global max pooling, and C is channel-wise concatenation.

For the high-level fusion, we directly concatenate the output features of the 4th blocks of the five branches (after the max pooling operation). Finally, we use a fully connected layer to generate the final prediction, giving the probability that the input sample belongs to which category (keratoconus, sub-clinical keratoconus, or normal).

In this specific structure, we take into account three major cues of prior knowledge. (1) Considering the access to the five slices of each sample, we expect that the five slices are correlated to one another; yet, the relation might not be as strong as that among the three channels in an RGB image. Therefore, five branches are constructed to respectively extract features of each slice, and then the low-level fusion and high-level fusion operations are performed to take into account the possible correlation among different slices. (2) Previous studies indicated that posterior elevation information is of great importance in clinical diagnosis of sub-clinical keratoconus [43], [48]. Thus, the middle branch is arranged for the image slice of posterior elevation (ELE-B), so that this branch can obtain more features from other slices and deliver more features to other branches. In this manner,

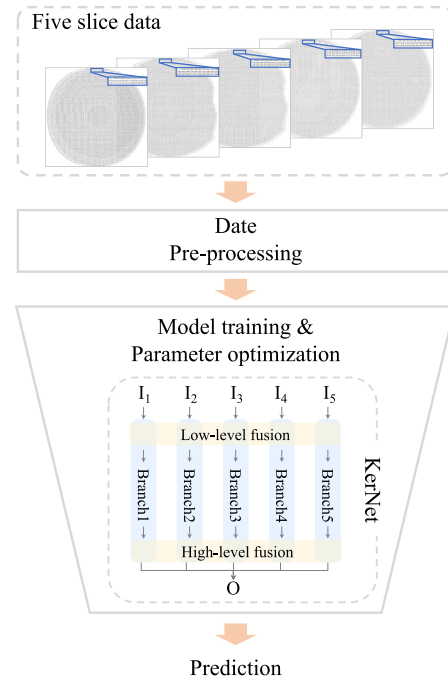


Fig. 5. The workflow of our approach.

the five slices are arranged in a more theoretically meaningful way to reduce the burden of a naive arrangement of building connections among the five branches. (3) Clinically, ophthalmologists generally pay more attention to certain regions of a slice, instead of paying the same amount of attention to the entire slice. Therefore, we introduce the spatial attention modules in the low-level fusion operation to guide our KerNet to work like a human analyzing the image slices. The experimental results suggest that this scheme of feature extraction and fusion yields better performance.

IV. EXPERIMENTS AND RESULTS

We conduct abundant experiments to demonstrate the effectiveness and superiority of our KerNet over other models. For fair comparison, we conduct all the experiments following the same workflow, as illustrated in Fig. 5. That is, pre-processing the collected dataset, training the model, and finally evaluating the model performance. This section follows such a pipeline. First, the detailed operations of the data pre-processing are presented. Next, the implementation details of model training (loss functions included) and the evaluation metrics are described. Finally, we present the experimental results and analyze the performance. It is worth noting that, since our KerNet aims for keratoconus and sub-clinical keratoconus detection, we measure the model performance based on its detecting ability for both the diseases. Also, we use the Class Activation Map (CAM) [49] to show where our KerNet focuses on, in order to link the feature extraction capability of KerNet with clinical practice.

A. Data Pre-Processing

As described in Section III-A2, we collected an in-house dataset containing 854 samples, including 335 normal samples, 172 sub-clinical keratoconus samples, and 347 keratoconus samples. For each sample, we obtain five numerical matrices of size 141×141 each (denoted as CUR-F, CUR-B, ELE-F, ELE-B, and PAC, respectively). Each numerical matrix is treated as a 2D image slice. That is, for each sample, five image slices, and a ground truth label L (indicating which category the sample belongs to), are available.

First, as done in the previous work (reviewed in Section III-A2), we crop all the matrices at their central region of size $5 \text{ mm} \times 5 \text{ mm}$ (about 101×101 elements). Next, we compute the mean values and standard deviation values for the five types of slices (i.e., CUR-F, CUR-B, ELE-F, ELE-B, and PAC) of all the samples, following the standard process of data standardization (e.g., as in [50]). Here, we show the example of computing the mean value and standard deviation value of CUR-F of the total N samples. By computing the mean value and standard deviation value of CUR-F of each sample (for the 101×101 elements), we obtain N mean values and N standard deviation values in total, corresponding to the N samples respectively. Then we average the N mean values to obtain the mean value of CUR-F and average the N standard deviation values to obtain the standard deviation value of CUR-F. The same operations are performed for the other four types of slices, and thus we obtain five mean values, [7.9, 6.46, 1068.87, 1105.1622], and five standard deviation values, [0.66, 0.76, 654.9672, 80.21], for CUR-F, CUR-B, ELE-F, ELE-B, and PAC, respectively, for all the samples. Also, taking CUR-F as an example to explain the operation of data normalization based on the aforementioned mean values and standard deviation values. We use the first element in the above mean value list and the first element in the standard deviation value list to conduct data normalization for the CUR-F slices of all the samples. The same normalization operations are performed for other types of slices for all the samples, respectively. For a more consistent performance, we split all the samples into a training set and a test set with a ratio of 8:2. Then, we split the samples of the training set into 5 groups to conduct a 5-fold cross validation, and finally test the models on the test set. For fairness of comparison, we conduct all the above pre-processing operations only once and use the dataset thus processed for all the experiments.

B. Experiments

1) **Implementation Details:** Our KerNet aims to detect keratoconus and sub-clinical keratoconus from samples, which can be regarded as a three-class classification task. It takes a sample of five image slices ($I_i, i \in \{1, 2, 3, 4, 5\}$) as input, each of size 101×101 , and predicts the probable category (P) of the sample. Specifically, the I_i 's, $i = 1, 2, \dots, 5$, correspond to CUR-B, CUR-F, ELE-B, ELE-F, and PAC. However, this classification task has to address the issue of imbalance data distribution for model training, as described in Section III-A2. The number of collected sub-clinical keratoconus samples in our dataset is only 172, nearly half of that of the keratoconus samples (347) and normal samples (335). Thus, to train KerNet,

TABLE I

THE LIST OF HYPER-PARAMETERS FOR KERNET AND ITS TRAINING

Hyper-parameters	Value
Input Size	101×101
Input Dimensions	5
Batch Size	16
Learning Rate	$1e-3$
Weight Decay Constant	$5e-4$
Max Iterations	50000

we use a Negative Log Likelihood (NLL) loss, a class-balanced cross-entropy loss function [51], which was shown to provide great benefit for classification tasks with an unbalanced training set [51]. The NLL loss function computes the difference between the predicted P and the ground truth L as follows:

$$\mathcal{L} = \sum_{j=1}^C W_j (-\alpha L \log(P) - (1 - \alpha)(1 - Y) \log(1 - P)) \quad (2)$$

where W_j represents a weight coefficient, which is assigned different values for a different class $j \in C$ ($C = 3$ in this task).

We set several hyper-parameters for our model KerNet, as listed in Table I. The first two are for data input, and the others are for model training. With an initial learning rate $1e-3$, we reduce it by a polynomial decay learning rate policy. We use the Adam optimizer with weight decay $5e-4$. We experimentally tried different combinations of the hyper-parameters for better performance, and the values in Table I yield the final performance in this work. Of course, employing a hyper-parameter optimization framework might give better results, but it is somewhat time-consuming. In this study, all the models were developed using Python 3.6.8 with the PyTorch deep learning library. All the experiments are constructed on an NVIDIA GeForce GTX 1080Ti GPU. The settings are identical for all the models, except for some hyper-parameters.

2) **Metrics:** We use multiple metrics to measure the classification performance of the models. First, we compute several metrics for measuring the overall performance, including the mean accuracy (ACC), mean recall (Recall), mean precision (Precision), F1 score (F1), and Kappa value (κ). All these metrics are computed based on the classification performance for the three categories (normal, keratoconus, and sub-clinical keratoconus). Given the True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) of all the categories, these metrics can be computed as:

$$ACC = \frac{TP + TN}{TP + FP + TN + FN} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (6)$$

Specifically, the Kappa value is a measure of agreement between the prediction P and ground truth label L . The higher its value is, the more accurate the model is. It can be computed as:

$$\kappa = (p_o - p_e) / (1 - p_e) \quad (7)$$

where p_o is the total classification accuracy (it can be computed by Eq. (3)), and p_e can be computed as $p_e = \frac{\sum_{i=1}^C a_i \times b_i}{n \times n}$, where C is the number of categories, n is the total number of samples, a_i is the number of expected samples and b_i is the number of predicted samples of class i .

Next, we further compute the accuracy values and F1 scores for the classes of keratoconus and sub-clinical keratoconus, respectively. These two metrics are used to evaluate the overall performance of the model on the target classes, including the precision and recall. For convenience, we denote these two metrics as ACC_k and $F1_k$ for keratoconus, and ACC_s and $F1_s$ for sub-clinical keratoconus.

Besides, we present the Receiver Operating Characteristic (ROC) curves and the AUC scores for the compared networks, to measure the classification performances. AUC is defined as the area of the ROC curve, which is defined by True Positive Rate (TPR) and False Positive Rate (FPR), respectively. The TPR and FPR are given by:

$$TPR = \frac{TP}{TP + FN} \quad (8)$$

$$FPR = \frac{FP}{FP + TN} \quad (9)$$

Thus, AUC can be computed by Eq. (10).

$$AUC = \int_{x=0}^1 TPR(FPR^{-1}(x))dx = P(X_1 > X_0) \quad (10)$$

where X_1 is the score for a positive instance and X_0 is the score for a negative instance.

C. Experimental Results

In this section, we first present the performance of our KerNet and explore the effectiveness of the specific structure and components of KerNet. Then, we demonstrate the superior performance of KerNet compared to other methods, from the perspectives of network structure and input data scheme. Finally, we use the Gradient-weighted Class Activation Mappings (Grad-CAM) [49] to show where our KerNet focuses on, which relates the model performance with clinical practice. All the compared models are implemented according to the original papers. All the parameters for the final test are selected on the basis of the best recall performance on the validation set.

1) Performance of KerNet: In this subsection, we present the performance of KerNet and investigate several possible operations of simply fusing features and adding additional attention modules under the premise of naive fusion. Here, we keep the operation of the final fusion (the final concatenation operation and fully connected layer) unchanged for all the attempted structures, since the network is expected to output one prediction

result. That is, the structure of the five separate branches with a high-level fusion operation is regarded as our baseline model. In this case, with four different possibilities of placing the feature fusion operations and two types of fusion choices (naive fusion operation and adding spatial attention modules), we conduct the experiments with a total of fifteen different structures. The first row of Table II gives a baseline performance and the third row shows the performance of our KerNet. The second to ninth rows present the performances of progressively adding more and more feature fusion operations. The tenth to fifteen rows are the performances of individually placing the feature fusion operations at different possible locations.

Obviously, the final structure of our KerNet yields slightly better performance (i.e., the results in the third row), reaching a good balance between the prediction accuracy and the number of parameters. We observe that (from the second to the ninth rows), not the more feature fusion operations we add to the baseline model, the better performance the model would give. This somewhat demonstrates the theoretical analysis given in Section III-B: Although both the low-level features and high-level features are expected to be fused, too many fusion operations will lead to redundancy (incurring additional computational costs but little performance improvement).

Besides, one may observe that among all the possible locations for applying the feature fusion operations, under the premise of naive fusion, adding additional spatial attention modules at the low-level locations further improves the performance noticeably (e.g., adding the spatial attention modules after the first Residual Block, as the second and third rows shown), but the same does not hold for the high-level locations. As shown in Table II, at the high-level locations, additional spatial attention modules may contribute little improvement (e.g., as shown in the eighth and the ninth rows, next-to-last row and the last row), or even lead to worse performance (e.g., as shown in the fourth and fifth rows, in the sixth and seventh rows, in the tenth and eleventh rows, and in the twelfth and the thirteenth rows). This is possibly due to that our spatial attention modules enable the network to pay more attention to spatial information. Note that it is well-known that high-level features tend to express more semantic information but lack distinct spatial characteristics [52].

2) Comparison With Other Methods: We then compare the performance of KerNet with other known methods, from two perspectives: (1) model structure and (2) input data scheme. First, we compare the performances of different model structures, including several representative image classification models, such as LeNet [53], AlexNet, VGGNet, ResNet18, ResNet50, DenseNet, and SE-ResNet. These models work in an end-to-end manner, receiving the five slices as input and predicting directly. It is worth noting that our KerNet takes five separate image slices as input (one slice per branch), while the other models receive the five slices as a whole (a 5-channel input). Table III shows the detailed results. For a more complete comparison, we present the Receiver Operating Characteristic (ROC) curves for these networks, as shown in Fig. 6. Obviously, for both keratoconus and sub-clinical keratoconus detection, KerNet achieves state-of-the-art performance nearly in all the

TABLE II

THE PERFORMANCE OF OUR KERNet AND THE VARIANTS USING DIFFERENT FEATURE FUSION OPTIONS (EITHER NAIVE FEATURE FUSION OPERATION OR ADDITIONAL SPATIAL ATTENTION MODULES) AT DIFFERENT LOCATIONS OF THE NETWORK. FOR CONVENIENCE, WE USE aR_i TO DENOTE ADDING FEATURE FUSION OPERATIONS AFTER THE i^{th} RESIDUAL BLOCK OF EACH BRANCH. USING NAIVE FEATURE FUSION OPERATION AND ADDING ADDITIONAL SPATIAL ATTENTION MODULES ARE RESPECTIVELY ABBREVIATED AS 'N' AND 'A'. KERATOCONUS AND SUB-CLINICAL KERATOCONUS ARE ABBREVIATED AS 'KC' AND 'SKC'. THE THIRD ROW GIVES THE PERFORMANCE OF KERNet

Locations				Options		Metrics (%)								
aR_1	aR_2	aR_3	aR_4	N	A	SKC		KC		Total				
						ACC_s	$F1_s$	ACC_k	$F1_k$	ACC	$Recall$	$Precision$	$F1$	κ
×	×	×	×	×	×	94.15	84.38	97.08	96.45	92.98	90.69	92.40	91.42	88.94
✓	×	×	×	✓	×	94.74	87.67	97.08	96.40	93.57	93.63	91.99	92.64	90.03
✓	×	×	×	✓	✓	95.91	91.18	98.25	98.57	94.74	93.71	94.10	93.89	91.78
✓	✓	×	×	✓	×	94.15	86.11	98.25	97.87	92.98	92.60	91.31	91.82	89.10
✓	✓	×	×	✓	✓	92.98	83.78	97.66	97.06	92.40	92.17	90.45	91.09	88.24
✓	✓	✓	×	✓	×	95.32	88.24	97.66	97.10	94.15	93.16	93.17	93.15	90.86
✓	✓	✓	×	✓	✓	94.74	86.57	97.66	97.18	93.57	92.13	92.55	92.33	89.92
✓	✓	✓	✓	✓	×	92.40	83.12	97.08	96.35	91.23	91.66	89.34	90.03	86.50
✓	✓	✓	✓	✓	✓	93.57	84.93	97.66	97.10	92.98	92.65	91.11	91.72	89.12
×	✓	×	×	✓	×	92.39	90.59	97.66	97.05	91.81	89.73	90.23	89.91	87.18
×	✓	×	×	✓	✓	91.22	81.01	98.24	97.81	90.64	91.13	88.67	89.23	85.65
×	×	✓	×	✓	×	91.81	80.55	98.24	97.81	91.22	90.18	89.07	89.53	86.38
×	×	✓	×	✓	✓	91.39	80.12	97.07	96.40	90.81	89.71	89.07	89.88	87.17
×	×	×	✓	✓	×	91.22	80.45	97.66	97.05	90.64	89.71	88.42	88.91	85.50
×	×	×	✓	✓	✓	91.56	82.05	98.24	97.84	91.98	90.65	90.31	90.47	88.04

TABLE III

COMPARISON OF OUR KERNet AND SEVERAL COMMONLY-USED MODELS. KERNet RECEIVES FIVE INDIVIDUAL SLICES OF A SAMPLE (ONE PER BRANCH), WHILE THE OTHER MODELS RECEIVE THESE SLICES AS A WHOLE (A 5-CHANNEL INPUT). KERATOCONUS AND SUB-CLINICAL KERATOCONUS ARE DENOTED AS 'KC' AND 'SKC,' RESPECTIVELY

Model	Metrics (%)								
	SKC		KC		Total				
	ACC_s	$F1_s$	ACC_k	$F1_k$	ACC	$Recall$	$Precision$	$F1$	κ
KerNet	95.91	91.18	98.25	98.57	94.74	93.71	94.10	93.89	91.78
LeNet	84.21	76.47	95.91	95.71	83.63	81.02	82.27	80.94	74.87
AlexNet	93.57	88.24	97.66	97.14	92.40	90.66	91.64	91.08	88.17
VGGNet	86.55	61.76	95.91	97.14	85.96	82.49	81.82	82.10	77.92
ResNet18	90.06	82.35	97.66	95.71	87.45	86.98	88.21	87.45	83.69
ResNet50	92.40	85.29	96.49	97.14	91.23	89.27	90.17	89.64	86.35
DenseNet	86.55	70.59	97.66	98.57	85.38	82.15	82.75	82.30	77.24
SE-ResNet	80.12	67.65	95.32	98.57	79.53	76.86	77.30	75.96	68.62
FFN [25]	85.96	68.42	97.08	96.40	84.80	83.26	81.89	82.24	76.55

metrics. Furthermore, KerNet shows notable superiority for sub-clinical keratoconus detection, although all the other models appear to perform well on keratoconus detection. For sub-clinical keratoconus detection, our KerNet outperforms all the other models by 3.4%-15.8% on ACC_s and 3%-13% on AUC , attaining significant improvement. Hence, KerNet provides a more effective method to distinguish sub-clinical keratoconus than other methods, which can alleviate the challenging issue that sub-clinical keratoconus samples are quite difficult to detect [3].

Then, we present the performances of FFN proposed in [25], which is one of the few existing approaches based on raw data of the Pentacam HR system (see the last row of Table III). The FFN, constructed following the original paper [25], conducts parameter selection in the first stage and makes final prediction in the second stage. Although the FFN obtains comparable performances with our KerNet for keratoconus detection, our KerNet shows noticeable superiority for sub-clinical keratoconus detection. Moreover, our KerNet works in an end-to-end manner,

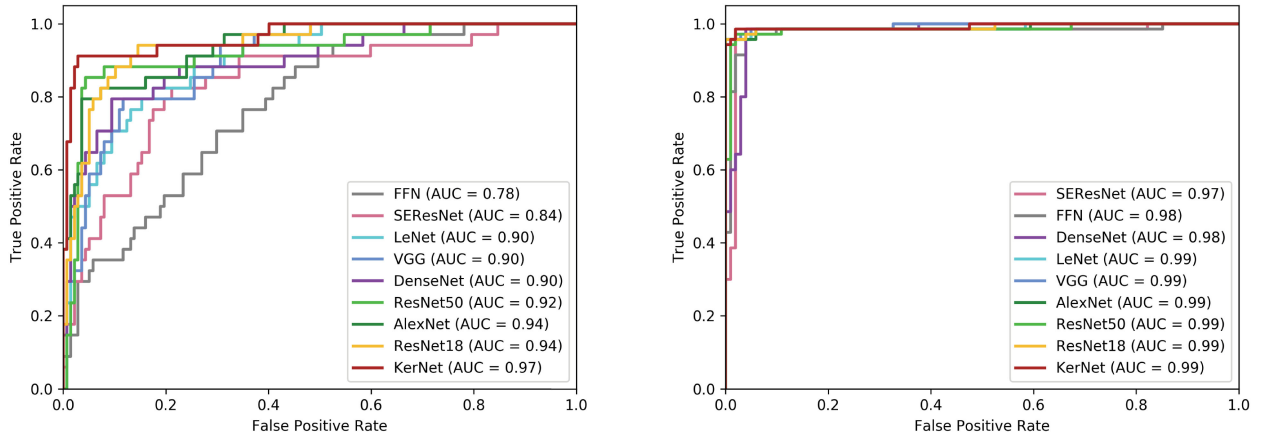


Fig. 6. The ROC curves of our KerNet and several commonly-used models. The left part shows the model performances of sub-clinical keratoconus detection and the right part shows the performances of keratoconus detection.

TABLE IV

COMPARISON OF DIFFERENT INPUT DATA SCHEMES, USING RESNET18 AS AN EXAMPLE. KERATOCONUS AND SUB-CLINICAL KERATOCONUS ARE DENOTED AS 'KC' AND 'SKC,' RESPECTIVELY

Input Data	Metrics (%)								
	SKC		KC		Total				
	ACC_s	$F1_s$	ACC_k	$F1_k$	ACC	$Recall$	$Precision$	$F1$	κ
ELE-B	88.30	65.52	96.49	95.71	87.72	82.37	86.00	83.37	80.45
CUR-B	87.72	66.67	95.91	95.10	86.55	82.32	83.69	82.85	78.77
CUR-F	84.21	65.82	96.49	95.59	84.21	82.81	81.53	81.63	75.79
PAC	83.04	60.27	93.57	92.62	81.87	78.80	78.73	78.19	71.85
ELE-F	76.61	58.33	94.15	92.96	75.44	76.29	77.14	72.87	63.38
Five Slices (Whole)	90.06	82.35	97.66	95.71	87.45	86.98	88.21	87.45	83.69
Five Slices (Separate)	94.15	84.38	97.08	96.45	92.98	90.69	92.40	91.42	88.94

which means that human interventions for feature selection are not needed.

We compare the effects of different schemes of input data. We simply use a ResNet18 to conduct the experiments, which can maintain a similar structure as KerNet. The experiments are conducted by feeding the ResNet18 with different numbers of image slices (one or five). Table IV shows the results. The first five rows give the results when we adapt ResNet18 for one-channel input and give it one slice at a time. The last two rows show the results when we adapt ResNet18 to receive all five slices simultaneously. The row before the last shows the results of a network receiving a whole five-channel input (*Five slices (Whole)*) for prediction. The last row is the results of a network with the same structure as that for the first row in Table II, which receives the five slices (*Five slices (Separate)*) using five separate branches and fuses the high-level features of each slice.

First, we observe that, with the same backbone structure, the model obtains better performance when it receives the entire five-slice information simultaneously, compared to receiving one of them at a time. Such an observation may follow from the experiments for both keratoconus detection and sub-clinical keratoconus detection. Second, the results of receiving five slices separately (the results in the last row of Table IV) are better than those of receiving them as a whole. Such results may support

our view that the correlation among the five slices is not as strong as that among the different channels of an RGB image (in Section III-B). Thus, simply treating the five slices as a whole leads to excessive fusion of features and possibly not very good performance. Of course, our specific structure of KerNet, moderately taking into account the possible correlation of the five slices, shows a better performance. The third observation from Table IV is that, when receiving one slice as input, the model receiving one ELE-B slice achieves a slightly better performance for sub-clinical keratoconus detection compared to the others. This further corroborates the previous research that posterior elevation profiles are of great importance in the diagnosis of sub-clinical keratoconus [43], [48], as mentioned in Section III-B.

Besides the noticable performance improvements of our KerNet, compared to the state-of-the-art networks (especially for sub-clinical keratoconus detection), it possibly incurs more memory and time costs. Our KerNet contains nearly 61.03 M parameters and makes prediction for one sample in around 0.02 seconds. As for AlexNet, which achieves the second best performances for sub-clinical keratoconus detection, it contains a similar amount of parameters (61.10 M) while makes prediction for one sample only in 0.001 seconds. For ResNet18, despite its weak performances, it is more light (with 11.708 M

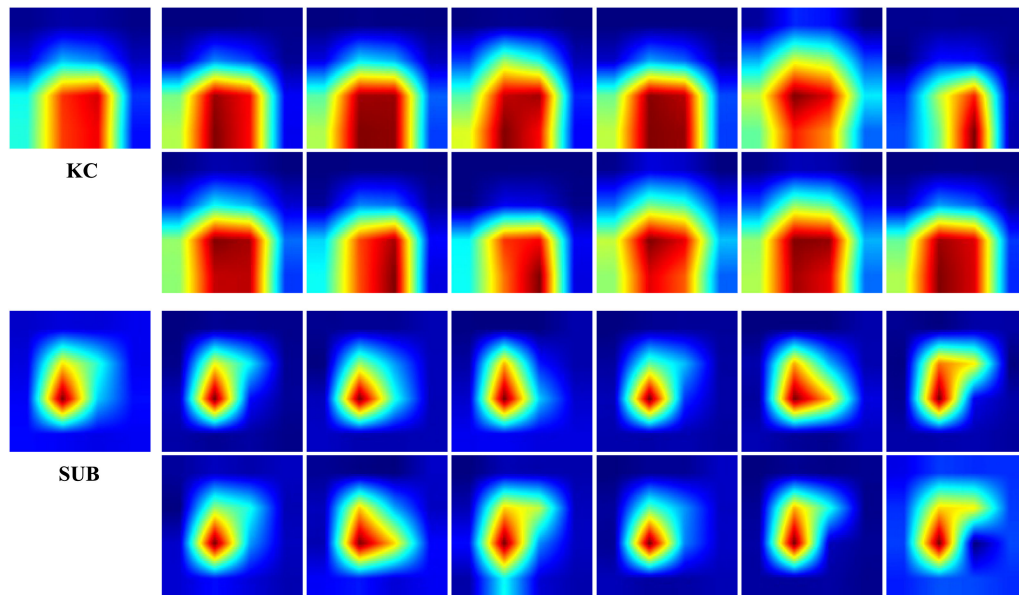


Fig. 7. Visualization of attention maps generated by Grad-CAM [49]. The top part shows the overall attention map for all the samples in *Group 1* (the upper left map) and the attention maps of 12 randomly selected samples in *Group 1* (i.e., samples correctly predicted as keratoconus). Similarly, the bottom part is for samples in *Group 2* (i.e., samples correctly predicted as sub-clinical keratoconus).

parameters) and can make more quick prediction (in 0.006 s). Facing the weaknesses of heavy memory and time costs, we will consider to achieve better trade-off between the costs and model performances in our future study.

3) Grad-CAM Visualization: Following the Gradient-weighted Class Activation Mappings (Grad-CAM) [49], we conduct an experiment to visualize where our KerNet chooses to look at in order to distinguish the keratoconus cases and sub-clinical keratoconus cases from the normal ones. For the target class, Grad-CAM uses the gradient of the final convolutional layer to produce an attention map, which highlights which areas of the image are important and class-discriminative in the detection of the target class. As for the specific tasks of medical image analysis, some researchers also used Grad-CAM for lesion detection, since the red regions corresponding to high scores of detecting disease samples possibly indicate the lesion areas [54], [55]. In this work, we adapt the Grad-CAM operation according to the specific structure of our KerNet, as illustrated in Fig. 8. First, the output feature maps of the last five convolutional layers of KerNet (five $512 \times 7 \times 7$ feature maps) are concatenated as a whole, which is regarded as the output feature maps of the “final convolutional layer” of KerNet, as in [49]. Then, Grad-CAM uses the gradient information flowing into the final convolutional layer to produce an attention map to highlight the important regions in the image for prediction. More detailed calculation of this process was explained in [49].

We use Grad-CAM to generate attention maps for all the samples that are predicted correctly as suffering diseases (i.e., the keratoconus and sub-clinical keratoconus samples), since our KerNet aims to detect those samples which show some possible lesion regions. The samples correctly predicted as keratoconus are denoted as *Group 1* and those correctly predicted as sub-clinical keratoconus are denoted as *Group 2*. For the samples in

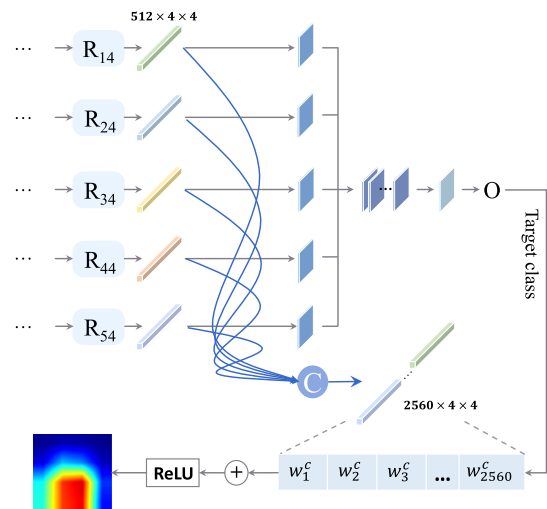


Fig. 8. Illustration of the specific Grad-CAM operation used in this study, following the work in [49].

Group 1, we average all the attention maps (in pixel-level) to obtain an overall attention map, which is the upper left map of the top part in Fig. 7. Such an overall attention map indicates to which places KerNet puts its attention when it distinguishes keratoconus samples from others (sub-clinical keratoconus and normal ones). The red regions correspond to high scores for keratoconus detection, indicating strong attention by KerNet. The blue regions represent low scores and low attention. Besides, 12 samples in *Group 1* are randomly selected and their attention maps are illustrated in the top part of Fig. 7 (to the right of the overall attention map). The same operations for the samples in *Group 2* and the attention maps are shown in the bottom part of Fig. 7. It is worth noting that the red regions (the focus areas of our KerNet) often occur in the lower left portions of the

attention maps, corresponding to the inferior temporal part of eyes (as discussed in Section III-A2). At the same time, the red regions of keratoconus samples are generally larger than those of sub-clinical keratoconus samples. These observations coincide with the previous studies that for keratoconus samples and sub-clinical keratoconus samples, 1) most cones were located in the inferior temporal quadrant [56], [57], and 2) keratoconus samples generally appear more serious lesions [58].

The above results have been verified by two independent experts. That is, our KerNet well notices the discriminative regions of keratoconus and sub-clinical keratoconus samples, which are in accordance with the human attention regions of such samples.

V. DISCUSSION AND CONCLUSION

In this paper, we presented a novel end-to-end deep learning approach, called KerNet, for keratoconus and sub-clinical keratoconus detection, using the raw data of the Pentacam HR system. The raw data exported from the Pentacam HR system is of a specific format, which consists of five slices for each sample (CUR-F, CUR-B, ELE-F, ELE-B, and PAC). Accordingly, we put five branches in our KerNet to receive the five slices separately, and introduced the idea of multi-level fusion (i.e., low-level fusion and high-level fusion) to moderately take into account the possible correlation among the five slices. Specifically, we introduced the attention mechanism, using spatial modules, to enhance the low-level fusion. For the high-level fusion, only simple channel-wise concatenation is conducted. Comparative evaluation experiments on our in-house dataset showed that our proposed KerNet can achieve far better results than state-of-the-art methods, especially for sub-clinical keratoconus detection. Besides, the attention maps generated by Grad-CAM showed that our KerNet places more attention on the inferior temporal part of the keratoconus and sub-clinical keratoconus samples (in accordance with clinical experts).

To our best knowledge, we are the first to propose an end-to-end deep learning approach utilizing raw data obtained by the Pentacam HR system for keratoconus and sub-clinical keratoconus detection. While our KerNet achieves high performances on detecting keratoconus and sub-clinical keratoconus, it possibly incurs more memory and time costs compared to some existing networks (as discussed in Section IV-C). Besides, it still cannot determine the sub-types or the severity of keratoconus diseases, which we will pursue in our future research.

REFERENCES

- [1] H. Hashemi *et al.*, "The prevalence and risk factors for keratoconus: A systematic review and meta-analysis," *Cornea*, vol. 39, no. 2, pp. 263–270, 2020.
- [2] N. Hallett *et al.*, "Deep learning based unsupervised and semi-supervised classification for keratoconus," 2020, *arXiv:2001.11653*.
- [3] D. Smadja *et al.*, "Detection of subclinical keratoconus using an automated decision tree classification," *Amer. J. Ophthalmol.*, vol. 156, no. 2, pp. 237–246, 2013.
- [4] F. Toutounchian, J. Shanbehzadeh, M. Khanlari, and A. Stage, "Detection of keratoconus and suspect keratoconus by machine vision," in *Proc. Int. Multi Conf. Eng. Comput. Scientists*, 2012, pp. 1–3.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [6] J. Du, W. Li, K. Lu, and B. Xiao, "An overview of multi-modal medical image fusion," *Neurocomputing*, vol. 215, pp. 3–20, 2016.
- [7] J. S. Velázquez-Blázquez *et al.*, "Detection of subclinical keratoconus using biometric parameters," in *Proc. Int. Work-Confer. Bioinf. Biomed. Eng.*, 2019, pp. 490–501.
- [8] I. A. N. Omar, "Keratoconus screening among myopic children," *Clin. Ophthalmol.* (Auckland, NZ), vol. 13, 2019, Art. no. 1909.
- [9] F. M. Elhennawi, Y. A. Alzankalony, M. K. Abdellatif, and A. M. Tawfik Ibrahim, "Role of anterior segment optical coherence tomography in the diagnosis of subclinical keratoconus in comparison with the pentacam," *Egyptian J. Hosp. Med.*, vol. 72, no. 1, pp. 3712–3715, 2018.
- [10] H. Hashemi *et al.*, "Pentacam accuracy in discriminating keratoconus from normal corneas: A diagnostic evaluation study," *Eye Contact Lens*, vol. 45, no. 1, pp. 46–50, 2019.
- [11] S. Yousefi *et al.*, "Keratoconus severity identification using unsupervised machine learning," *PLoS One*, vol. 13, no. 11, 2018, Art. no. e0205998.
- [12] M. Malfeito, U. Regueiro, M. Pérez-Mato, F. Campos, T. Sobrino, and I. Lema, "Innate immunity biomarkers for early detection of keratoconus," *Ocular Immunol. Inflammation*, vol. 27, no. 6, pp. 942–948, 2019.
- [13] I. Kovács *et al.*, "Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus," *J. Cataract Refractive Surg.*, vol. 42, no. 2, pp. 275–283, 2016.
- [14] A. Lavric and P. Valentin, "Keratodetect: Keratoconus detection algorithm using convolutional neural networks," *Comput. Intell. Neurosci.*, vol. 2019, 2019, pp. 1–9.
- [15] S. Huseynli, J. Salgado-Borges, and J. L. Alio, "Comparative evaluation of scheimpflug tomography parameters between thin non-keratoconic, sub-clinical keratoconic, and mild keratoconic corneas," *Eur. J. Ophthalmol.*, vol. 28, no. 5, pp. 521–534, 2018.
- [16] H. Hashemi *et al.*, "Keratoconus indices and their determinants in healthy eyes of a rural population," *J. Curr. Ophthalmol.*, 2019.
- [17] M. Koc *et al.*, "Topometric and tomographic evaluation of subclinical keratoconus," *Ophthalmic Epidemiol.*, vol. 17, no. 4, pp. 1–9, 2020.
- [18] E. Atalay, O. Özalp, M. A. Erol, M. Bilgin, and N. Yildirim, "A combined biomechanical and tomographic model for identifying cases of subclinical keratoconus," *Cornea*, vol. 39, no. 4, pp. 461–467, 2020.
- [19] H. Zou, J. Xu, L. Zhang, S. Ji, and Y. Wang, "Assistant diagnose for subclinical keratoconus by artificial intelligence," *Chin. J. Ophthalmol.*, vol. 55, no. 12, p. 911–915, 2019.
- [20] B. T. Lopes *et al.*, "Enhanced tomographic assessment to detect corneal ectasia based on artificial intelligence," *Amer. J. Ophthalmol.*, vol. 195, pp. 223–232, 2018.
- [21] I. R. Hidalgo *et al.*, "Evaluation of a machine-learning classifier for keratoconus detection based on scheimpflug tomography," *Cornea*, vol. 35, no. 6, pp. 827–832, 2016.
- [22] H. Hashemi, A. Beiranvand, A. Yekta, A. Maleki, N. Yazdani, and M. Khabazkhoob, "Pentacam top indices for diagnosing subclinical and definite keratoconus," *J. Curr. Ophthalmol.*, vol. 28, no. 1, pp. 21–26, 2016.
- [23] D. T. Hogarty, D. A. Mackey, and A. W. Hewitt, "Current state and future prospects of artificial intelligence in ophthalmology: A review," *Clin. Exp. Ophthalmol.*, vol. 47, no. 1, pp. 128–139, 2019.
- [24] Y. Xie *et al.*, "Screening candidates for refractive surgery with corneal tomographic-based deep learning," *JAMA Ophthalmol.*, vol. 138, no. 5, pp. 519–526, 2020.
- [25] I. Issarti, A. Consejo, M. Jiménez-García, S. Hershko, C. Koppen, and J. J. Rozema, "Computer aided diagnosis for suspect keratoconus detection," *Comput. Biol. Med.*, vol. 109, pp. 33–42, 2019.
- [26] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
- [27] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [28] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 1–9.
- [29] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4700–4708.
- [30] A. Kumar, J. Kim, D. Lyndon, M. Fulham, and D. Feng, "An ensemble of fine-tuned convolutional neural networks for medical image classification," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 1, pp. 31–40, Jan. 2017.

- [31] I. P. Okuwobi, Z. Ji, W. Fan, S. Yuan, L. Bekalo, and Q. Chen, "Automated quantification of hyperreflective foci in SD-OCT with diabetic retinopathy," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 4, pp. 1125–1136, Apr. 2020.
- [32] P. Burlina, D. E. Freund, N. Joshi, Y. Wolfson, and N. M. Bressler, "Detection of age-related macular degeneration via deep learning," in *Proc. IEEE 13th Int. Symp. Biomed. Imag.*, 2016, pp. 184–188.
- [33] X. Xu, L. Zhang, J. Li, Y. Guan, and L. Zhang, "A hybrid global-local representation CNN model for automatic cataract grading," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 2, pp. 556–567, Feb. 2020.
- [34] G. W. Armstrong and A. C. Lorch, "A(eye): A review of current applications of artificial intelligence and machine learning in ophthalmology," *Int. Ophthalmol. Clin.*, vol. 60, no. 1, pp. 57–71, 2020.
- [35] D. S. J. Ting *et al.*, "Artificial intelligence for anterior segment diseases: Emerging applications in ophthalmology," *Brit. J. Ophthalmol.*, vol. 105, no. 2, pp. 158–168, 2021.
- [36] A. Lavric, V. Popa, C. David, and C. C. Paval, "Keratoconus detection algorithm using convolutional neural networks: Challenges," in *Proc. 11th Int. Conf. Electron., Comput. Artif. Intell.*, 2019, pp. 1–4.
- [37] K. Kamiya *et al.*, "Keratoconus detection using deep learning of colour-coded maps with anterior segment optical coherence tomography: A diagnostic accuracy study," *BMJ Open*, vol. 9, no. 9, 2019, Art. no. e031313.
- [38] V. A. Dos Santos *et al.*, "CorneaNet: Fast segmentation of cornea OCT scans of healthy and keratoconic eyes using deep learning," *Biomed. Opt. Exp.*, vol. 10, no. 2, pp. 622–641, 2019.
- [39] R. H. Silverman, R. Urs, A. RoyChoudhury, T. J. Archer, M. Gobbe, and D. Z. Reinstein, "Epithelial remodeling as basis for machine-based identification of keratoconus," *Invest. Ophthalmol. Vis. Sci.*, vol. 55, no. 3, pp. 1580–1587, 2014.
- [40] P. A. Accardo and S. Pensiero, "Neural network-based system for early keratoconus detection from corneal topography," *J. Biomed. Informat.*, vol. 35, no. 3, pp. 151–159, 2002.
- [41] P. Zéboulon, G. Debellemanière, M. Bouvet, and D. Gatinel, "Corneal topography raw data classification using a convolutional neural network," *Amer. J. Ophthalmol.*, vol. 219, pp. 33–39, 2020.
- [42] U. de Sanctis, C. Loiacono, L. Richiardi, D. Turco, B. Mutani, and F. M. Grignolo, "Sensitivity and specificity of posterior corneal elevation measured by pentacam in discriminating keratoconus/subclinical keratoconus," *Ophthalmology*, vol. 115, no. 9, pp. 1534–1539, 2008.
- [43] Z. Xu *et al.*, "Value of corneal epithelial and bowman's layer vertical thickness profiles generated by UHR-OCT for sub-clinical keratoconus diagnosis," *Sci. Rep.*, vol. 6, no. 1, pp. 1–11, 2016.
- [44] R. Feng *et al.*, "SSN: A stair-shape network for real-time polyp segmentation in colonoscopy images," in *Proc. IEEE 17th Int. Symp. Biomed. Imag.*, 2020, pp. 225–229.
- [45] J. Bühren, D. Kook, G. Yoon, and T. Kohnen, "Detection of subclinical keratoconus by using corneal anterior and posterior surface aberrations and thickness spatial profiles," *Invest. Ophthalmol. Vis. Sci.*, vol. 51, no. 7, pp. 3424–3432, 2010.
- [46] Z. Zhang, X. Zhang, C. Peng, X. Xue, and J. Sun, "ExFuse: Enhancing feature fusion for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 269–284.
- [47] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
- [48] M. A. Shousha *et al.*, "The use of Bowman's layer vertical topographic thickness map in the diagnosis of keratoconus," *Ophthalmology*, vol. 121, no. 5, pp. 988–993, 2014.
- [49] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 618–626.
- [50] O. Tasar, Y. Tarabalka, A. Giros, P. Alliez, and S. Clerc, "Standard-GAN: Multi-source domain adaptation for semantic segmentation of very high resolution satellite images by data standardization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2020, pp. 192–193.
- [51] J. T. Barron, "A general and adaptive robust loss function," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4331–4339.
- [52] L.-J. Li, H. Su, L. Fei-Fei, and E. P. Xing, "Object bank: A high-level image representation for scene classification & semantic feature sparsification," in *Proc. Adv. Neural Inf. Process. Syst.*, 2010, pp. 1378–1386.
- [53] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [54] H. Panwar, P. Gupta, M. K. Siddiqui, R. Morales-Menendez, P. Bhardwaj, and V. Singh, "A deep learning and grad-CAM based color visualization approach for fast detection of COVID-19 cases using chest X-ray and CT-scan images," *Chaos, Solitons Fractals*, vol. 140, 2020, Art. no. 110190.
- [55] H. Jiang *et al.*, "A multi-label deep learning model with interpretable Grad-CAM for diabetic retinopathy classification," in *Proc. 42nd IEEE Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2020, pp. 1560–1563.
- [56] Z. Zhang *et al.*, "Corneal asphericity and its related factors in 1052 chinese subjects," *Optometry Vis. Sci.*, vol. 88, no. 10, pp. 1232–1239, 2011.
- [57] G. U. Auffarth, L. Wang, and H. E. Völcker, "Keratoconus evaluation using the orbscan topography system," *J. Cataract Refractive Surg.*, vol. 26, no. 2, pp. 222–228, 2000.
- [58] J. C. Reddy, C. J. Rapuano, J. R. Cater, K. Suri, P. K. Nagra, and K. M. Hammersmith, "Comparative evaluation of dual Scheimpflug imaging parameters in keratoconus, early keratoconus, and normal eyes," *J. Cataract Refractive Surg.*, vol. 40, no. 4, pp. 582–592, 2014.